

Can ProteinGym scores support a model decision?

Power, rank uncertainty and assay-design sensitivity on the zero-shot DMS benchmark

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Decisions about AI models in biology rest on evaluation scores. On the field’s main public protein benchmark, the smallest gain reliably detectable at the top is about 0.015 (80% power), larger than the gap between the published #1 and #2 and comparable to the gap to #3. Which model ranks #1 also depends on which kinds of lab tests are counted—a design choice, not a property of the models. We ask how large an evaluation must be before its numbers can support a decision. **Summary.** At commit 144fe22, 97 models and 217 assays, today’s benchmark could support claims about gains of roughly 0.020 Spearman or more over the leader (0.015–0.027 across ten pairs); $\Delta = 0.01$ would need about 850 assays (3.9 $\times$ today). Rank intervals leave 3 models not ruled out as #1; only 2 of 19 adjacent top-20 pairs differ significantly. Shared-assay re-ranking makes VenusREM #1 on 200 assays (vs AIDO Protein-RAG on all assays). We treat leaderboard scores as a *hypothesis* of preferability for mutation-effect prediction, not evidence of dangerous capability.

1 Question

ProteinGym [12] reports bootstrap SEs of each model’s gap to #1, not rank uncertainty. Neuhoof and Benjamini [11] give rank intervals for TabArena and MMLU; we apply that style of max- t inference and power analysis to ProteinGym’s nested structure. Woolley et al. [14] warn that aggregates hide dataset-level variance; Livesey and Marsh [7, 8] bootstrap rank significance on their own DMS benchmarks. To our knowledge, this is the first simultaneous rank-interval and power analysis of the ProteinGym DMS leaderboard with assay-subset sensitivity; we do not claim the first rank intervals ever [4, 11].

2 Data and version

We use ProteinGym GitHub commit 144fe22 (25 March 2026): release v1.3 plus AIDO Protein-RAG and ProtRiever, 97 models and 217 assays. As in ProteinGym’s script, assay scores are averaged within (UniProt ID, function) units (200 units), units within five function groups (Activity, Binding, Expression, OrganismalFitness, Stability), and the score is the mean of the five group means. We reproduce 97/97 averages and 97/97 ranks; 91/97 SEs match at 3 dp. We report overall statistics only and do not stratify by organism or viral proteins, as a biosecurity-adjacent judgment rather than a data limitation.

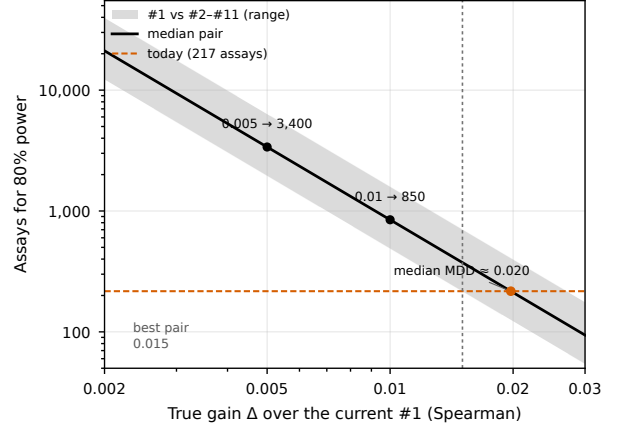
3 Methods

Estimand. Let θ_m be model m ’s ProteinGym aggregate in the limit of infinitely many units per function group, treating the benchmark’s units as a sample from each group. Its rank is $r_m = 1 + \#\{k : \theta_k > \theta_m\}$. Our intervals quantify the uncertainty due to which proteins happen to be in the benchmark. They are not prediction intervals for a model’s rank on a *new* assay, which is Neuhoof and Benjamini’s target and is much wider.

Bootstrap. Exactly as for ProteinGym’s error bars, units are re-sampled with replacement within function groups ($B = 10,000$), a stratified bootstrap in the sense of [1]. Missing scores are skipped, as in ProteinGym.

Rank intervals. Following Mogstad et al. [10], let $T_{jk} = (\hat{\theta}_j - \hat{\theta}_k - \theta_j + \theta_k) / \widehat{se}_{jk}$. The *marginal* interval for model j takes c_j , the 95% bootstrap quantile of $\max_k |T_{jk}^*|$, and sets the lower bound to $1 + \#\{k : \hat{\theta}_k - \hat{\theta}_j > c_j \widehat{se}_{jk}\}$ and the upper bound to $p - \#\{k : \hat{\theta}_j - \hat{\theta}_k > c_j \widehat{se}_{jk}\}$; it covers r_j with probability at least 95% for each j . The *simultaneous* interval uses one critical value from $\max_{j,k} |T_{jk}^*|$ and covers all ranks jointly; its step-down version [13] recomputes the critical value over the comparisons not yet rejected. The 95% best-model set keeps j unless some model beats it at a one-sided max- t critical value, so it contains the true #1 with probability at least 95%. We report marginal single-step and simultaneous step-down intervals. The percentile interval of bootstrap ranks is shown for contrast; it has no guarantee and fails near ties [2, 5]. Adjacent pairs in the top 20 get paired bootstrap z -tests with Holm’s correction [6].

a Does this gain support a decision?
(80% power, paired test, $\alpha = 0.05$)



b Supporting rank CIs (top 7; 3 not ruled out as #1)

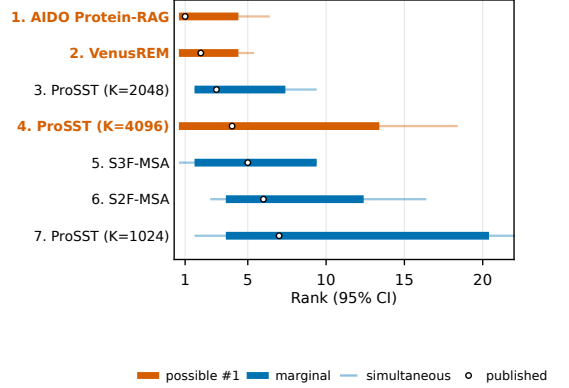


Figure 1: (a) Assays for 80% power to detect gain Δ over the current #1 (band: #1 vs next ten). Today: 217 assays; median MDD ≈ 0.020 ; best pair ≈ 0.015 . **(b)** Supporting rank CIs for top seven (marginal thick, simultaneous thin); orange = not ruled out as #1.

Power. For #1 and each of the next ten complete-coverage models we compute the per-unit paired differences and the standard error $\widehat{se}_0$ of their group-balanced mean. A true gain Δ is detected with 80% power by a two-sided test at $\alpha = 0.05$ with $n = n_0 [(\widehat{z}_{0.975} + \widehat{z}_{0.8}) \widehat{se}_0 / \Delta]^2$ units, assuming new assays arrive as new units in today’s function mix (217/200 assays per unit) [3, 9]. We check the formula by resampling the real paired differences re-centred at Δ .

Coverage simulation. Five scenarios, 1,000 synthetic leaderboards each (1,000 bootstrap replicates), as in the repository; tied ranks require set coverage [11].

4 Results

Power (Fig. 1a). With 80% power at today’s 200 units, the median MDD over #1 vs the next ten is 0.020 (0.015–0.027); the best pair is 0.015, above the 0.00004 gap between AIDO Protein-RAG and VenusREM and near the 0.011 gap to #3. $\Delta = 0.01$ needs ≈ 850 assays (3.9 $\times$); $\Delta = 0.005$ needs $\approx 3,380$ (15.6 $\times$). We phrase “this gap supports a decision” as a hypothesis at stated Δ , not a #1 verdict.

(A) Statistical uncertainty (Fig. 1b). AIDO Protein-RAG and VenusREM each lead in $\approx 49\%$ and 50% of bootstrap draws; the 95% best-model set is AIDO Protein-RAG, VenusREM and ProSST (K=4096). Only 2 adjacent top-20 pairs differ significantly. Marginal intervals keep worst-model coverage ≥ 0.973 in simulation; simulta-

Table 1: Shared-assay re-ranking (`shared_assay_rerank.csv`).

Subset	n assays	#1 / possible #1s
All assays	217	AIDO Protein-RAG / AIDO Protein-RAG, VenusREM and ProSST (K=4096)
All models scored	200	VenusREM / ProSST (K=4096), VenusREM and AIDO Protein-RAG
Top 10 scored	200	VenusREM / ProSST (K=4096), VenusREM and AIDO Protein-RAG

neous step-down ≥ 0.994 ; naive percentile ranks drop to 0.081 under exact ties.

(B) Assay mix and averaging (design). Dropping Activity or OrganismalFitness makes VenusREM #1. On all 200 assays scored by every model, VenusREM leads (vs AIDO Protein-RAG on all 217); the possible-#1 set is ProSST (K=4096), VenusREM and AIDO Protein-RAG (Table 1). The published top- $K=10$ share the same 200 assays here. NDCG reorders sharply (4/10 of the published top 10 remain). Prototriever (#8) falls to #10 on shared assays; its missing assays average 0.379 vs 0.410 for others.

5 Limitations

Intervals omit mutant-level noise; power assumes future assays resemble today’s mix. High mutation-effect scores are not evidence of misuse potential. Literature search 6 October 2026.

References

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